

Predicting Diabetes Risk with Health-Survey Indicators: Classifying Classical Machine Learning to Deep Learning.

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- GitHub repository: [Summative Repo](#)
- Demo video: [INSERT LINK]

Abstract

Diabetes prevalence is growing most rapidly in areas of low- and middle-income, most of which suffer from poor laboratory-based screening, and where most diabetes cases are undiagnosed. This project aims to determine the predictiveness of diabetes or pre-diabetes based on the low-cost survey data that could be easily collected, and to see if classical machine learning or deep learning is best for the task. This data set contains 253,680 rows of data and 21 features representing health indicators from the CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015, so I created a single leak-free preprocessing pipeline and conducted nine experiments, ranging from logistic regression, random forests, histogram gradient boosting, and a support vector machine on the classical side, to Sequential and Functional TensorFlow networks on the deep-learning side. The ROC-AUC was statistically similar across all nine models, ranging from 0.78 to 0.82, and the best deep network (0.818) and gradient boosting (0.817). Class weighting and decision-threshold tuning could improve recall from 0.17 to 0.61, but had no impact on model ranking, which shows that the binding constraint is the decision threshold, not the model capacity. A detailed inspection of errors within subgroups showed that the overall accuracy masks a socio-economic gap: the higher the income and the higher the level of education, the lower the recall on the true cases, and thus a single threshold for the total error in the test yields systematically different error profiles for the subgroups. I believe the most pragmatic one is gradient boosting, and that fairness should be the primary engineering challenge when it comes to any practical screening application.

1. Introduction

Diabetes mellitus is no longer just a disease of rich nations, but has become one of the most important NCD burdens in developing countries. It is estimated that in 2024, there were approximately 589 million adults aged 20 to 79 living with diabetes and this number is expected to rise sharply over the next few decades [1]. Sub-Saharan Africa (SSA) is projected to see the highest relative increase in diabetes cases, with the number of people living with diabetes projected to grow by 142 percent by 2050, the largest increase among the regions, due to the high rate of urbanisation in the region, ageing populations, dietary shift and increasing obesity [1] and [2]. What is dangerous about this trajectory is that nearly three-quarters of adults with diabetes in the African region are undiagnosed (the highest undiagnosed rate of any region), with low spending on diabetes-related health services [1]. Where laboratory confirmation is expensive and scarce, like Rwanda, many patients go untreated until symptoms of complications develop.

This is where low cost data-driven screening becomes valuable and complements the rising burden and lack of diagnostic capacity and a hidden majority. The model cannot diagnose a patient but can be used as a triage tool to identify the patients who need to have a confirmatory test and be used where it

will help, meaning that limited laboratory resources are used where they are needed. The question this project is addressing is a practical one: What is the best prediction of diabetes or pre-diabetes status possible, using behavioural, clinical and socio-economic measures routinely collected, and what family of machine-learning methods does this best?

For this study, I've used the Behavioral Risk Factor Surveillance System (BRFSS) 2015 diabetes health-indicators dataset, which has a sample of 253,680 respondents to the survey, with each person having 21 recorded features including clinical history (high blood pressure, high cholesterol, prior stroke or heart disease); lifestyle (physical activity, diet, smoking, alcohol use); self-rated health; and demographics (age, sex, education, income) [3]. There are two properties of the data set that influence the overall methodology. First, it is highly imbalanced (only around 1 respondent out of 7 gives a positive diagnosis), making raw accuracy a poor measure of success, and it encourages measures based on ordering and thresholds. Secondly, survey data is cardinality low, which means there are many duplicate records in the data, and these need to be eliminated before splitting otherwise there could be optimistic contamination between the training partition and the test partition.

Classical machine learning and deep learning are compared in a controlled setting using the same data and the same data pre-processing. On the classical side I consider logistic regression, random forests, histogram based gradient boosting and a support vector machine. On the deep-learning side I use a Sequential multilayer perceptron, a regularised version of it with batch normalisation and dropout and a two-branch Functional model, all implemented with the TensorFlow tf.data input pipeline. Each model is evaluated using one evaluation harness for the nine experiments to be directly comparable. This work has two fold value. Conceptually, it offers a classic versus deep comparison that is like-for-like on a large tabular health datasets, and applies the now substantial body of research that suggests that tree-based models still have an edge on tabular problems to real data. Substantively and more significantly, it goes beyond just one level of accuracy and inquires as to who it misses. The degree of identifying as true cases is not uniform across income and education bands of the population, however, and this is obscured completely by the overall model accuracy of about 0.85 that is reported in the headline. The rest of the report is a review of the literature, a description of the data and methods, presentation of experiments and analysis of errors, and discussion of the implications and limitations of using such a model in a resource limited environment.

2. Literature Review

2.1: Machine learning for diabetes risk prediction.

The prediction of diabetes using survey and electronic-health-record data is a well-studied problem and the BRFSS health-indicators data set has recently been used as a common benchmark. Typically, reported studies use a mix of classifiers, including logistic regression, random forest and gradient boosting (such as XGBoost or CatBoost), and achieve excellent overall results, with boosted ensembles often being the best-performing, and body-mass index, high blood pressure, and cholesterol, as well as age, consistently appearing as the strongest predictors [12]–[13]. This collection of research confirms the existence of the predictive signal within BRFSS data and the effectiveness of ensemble techniques to capitalize on this signal. However, its lack of telling data is limited to aggregate metrics. Very few studies ask about the location of the remaining errors, and even fewer about whether the predictive quality of a survey is equal across the socio-economic strata

measured in the survey. The gap the present project seeks to address is not a mere incremental accuracy comparison, but a clear analysis of the performance in a data set where income and education are provided to allow for such analysis.

2.2 Classical versus deep learning on tabular data.

One key question of design for this project is if the representational ability of neural networks is correlated with the performance of the network on structured survey data. This is a point that is consistently found in recent literature. Having a much lower computational cost, Grinsztajn and colleagues compared tree-based models with a variety of standard and novel deep models over forty-five datasets and a very large hyperparameter search, and the tree-based models were still the state of the art on medium sized tabular problems [4]. They believe that this disparity is due to the different inductive biases of neural networks: a neural network tends to be sensitive to the orientation of features, and tends to be hard to train when the target function is irregular, whereas decision trees are not sensitive to the orientation of the features. This difference is explained by the different inductive biases of neural networks - they are sensitive to the orientation of features, and they are harder to train when the target function is irregular, while decision trees do not have this. Shwartz-Ziv and Armon were able to come up with a similar conclusion: specialized deep models did not necessarily outperform well-tuned gradient-boosted trees, and often did not, while requiring more tuning effort than these models [5]. A more general survey by Borisov and coworkers also considers tabular deep learning an open problem, not a solved problem [6]. So my bias going into the experiments was that nothing would replace deep learning, but that gradient boosting would be at least as good as deep learning, but would be cheaper, and more interpretable, which is what I saw. This tabular strength is the reason why the foundational methods behind the classical side of machine learning, like gradient boosting [7] and its scalable boosted-tree implementation [8] are so well known.

2.3 Learning from imbalanced data

As positive cases are a minority in the BRFSS data, this project falls right in the middle of the imbalanced-learning literature. Synthetic minority over-sampling technique (SMOTE) proposed by Chawla and his co-authors is a standard baseline at the data level, which creates a minority class by interpolating new minority instances [9]. He and Garcia's survey catalogues a much wider range of remedies for imbalance, at the level of accuracy, cost-sensitive, and algorithm-level; and, importantly in the context of this work, argues that accuracy is not an appropriate objective under imbalance, favoring instead the areas under the ROC and precision/recall curves [10]. One common observation is that failure of a classifier on the minority class does not necessarily indicate failure of the classifier, but failure to tune the default decision threshold for the minority class. This definition is directly used in my methodology: the imbalance is not just a modelling problem to be addressed by either making larger models or rebalancing models, but by using class weights and then tuning the decision threshold on validation data, reporting a ROC-AUC score (the area under the precision-recall curve, as well as F1 scores, recall and precision scores, and the decision threshold).

2.4. Prediction models for health: fairness and equity

The most significant piece of work related to algorithmic equity in health is the work of the Absent Health Alliance. The Absent Health Alliance is the most impactful strand of related work. However, Obermeyer and colleagues have shown that such a widely used commercial risk-prediction algorithm, which impacts millions of patients, systematically over-identified Black patients for supplemental care, as it was predicting health-care cost as an indicator of the health need, and that unequal access resulted in less health-care spending on similarly sick Black patients [11]. The key point is that a model might seem correct given certain summary statistics, yet have significant differences in others, and that these differences might actually be due to the nature of the data and labels themselves, not necessarily to the individual errors [11]. The same because of the risk here. BRFSS label is a self-reported diagnosis and diagnosis relies on access to care, which has a gradient based on income and education. A well-performing model overall can thus have very different behaviour across socio-economic groups. This is why I view the disaggregation by sex, education level, income and age as a key outcome, not an as-required or as-optional result.

2.5 Synthesis and positioning

The literature, combined, outlines three expectations and one gap. First, the diabetes-prediction work confirms that BRFSS has a strong and learnable signal, but is overwhelmingly measured on the aggregate level [12, 13]. Second, the tabular-learning literature suggests that gradient boosting will be as effective at this type of data as neural networks and is cheaper and more interpretable [4, 5, 6]. Third, the literature on imbalanced learning frames the minority class problem as thresholding and metric choice, instead of capacity [9, 10]. The gap, and that of this project, is between the first and fourth strands; the diabetes-prediction task is evaluated by the equity lens of Obermeyer and colleagues [11] by comparing the rank of the cases not just by performance overall but by performance across the socio-economic groups that a deployed screening tool would serve.

3. Methodology

3.1 Dataset and exploratory analysis

There are 253,680 BRFSS 2015 respondents and 21 predictors along with a binary target column Diabetes_binary (self-reported diabetes or pre-diabetes) [3]. The predictors can be categorized into 3 categories. Clinical markers are the blood pressure, cholesterol, recent cholesterol check, prior stroke, prior heart disease, and difficulty walking and BMI. Behavioural factors encompass physical activity, eating of fruit and vegetables, smoking and high alcohol consumption. The rest of the fields are self-rated general, mental and physical health along with the following demographics: age band, sex, education level and income band. There are no missing values in the data.

Three facts arose in exploratory analysis that were the basis for all decisions made later. The first problem is that the target is imbalanced - only 13.9% of the raw respondents have a positive diagnosis - which means that any model that simply guesses the majority class would be the most accurate. Second, a number of predictors have a strong and clinically sensible signal. Each of the classes is clearly separated with respect to self-rated general health, high blood pressure, high cholesterol, BMI, difficulty walking and age, and there is a clear socio-economic gradient in which the prevalence of diabetes increases with decreasing income and education. Third, the survey does not have any missing values but there are 24,206 exact duplicates, that is, identical answer vectors that occur naturally in the survey due to low cardinality of the features; there are also real high outliers in the body-mass index. These findings are summarised in figures 1-3.

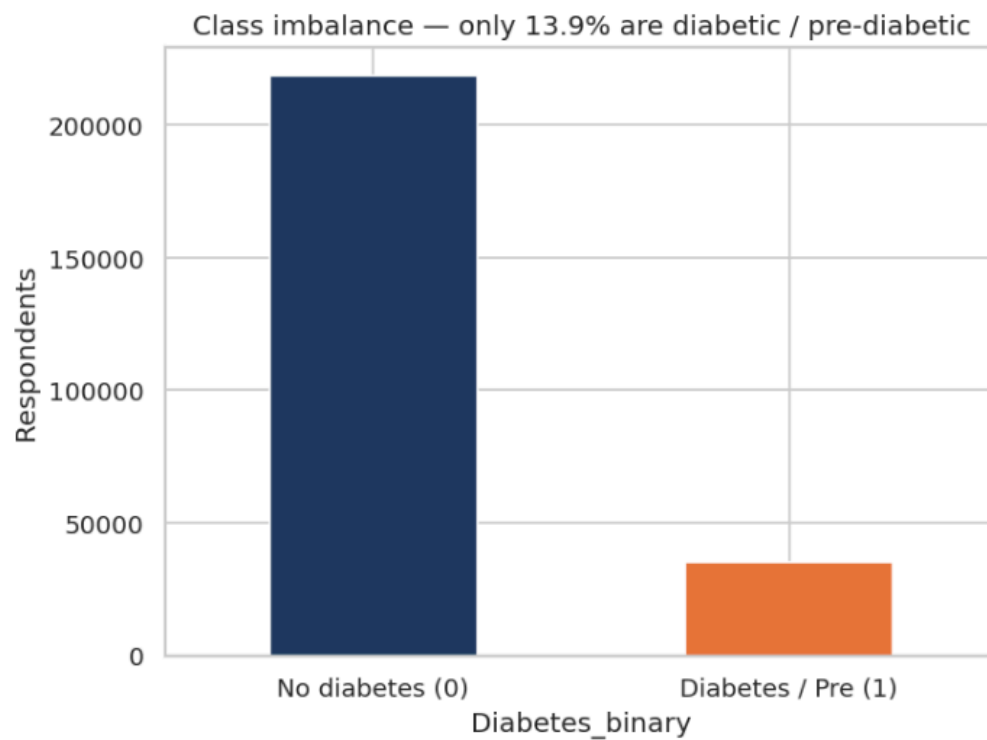


Figure 1. Class distribution. The number of positive cases (cases with diabetes, or pre-diabetes) is a clear minority, and motivates the ranking-based and threshold-aware measures, rather than accuracy.

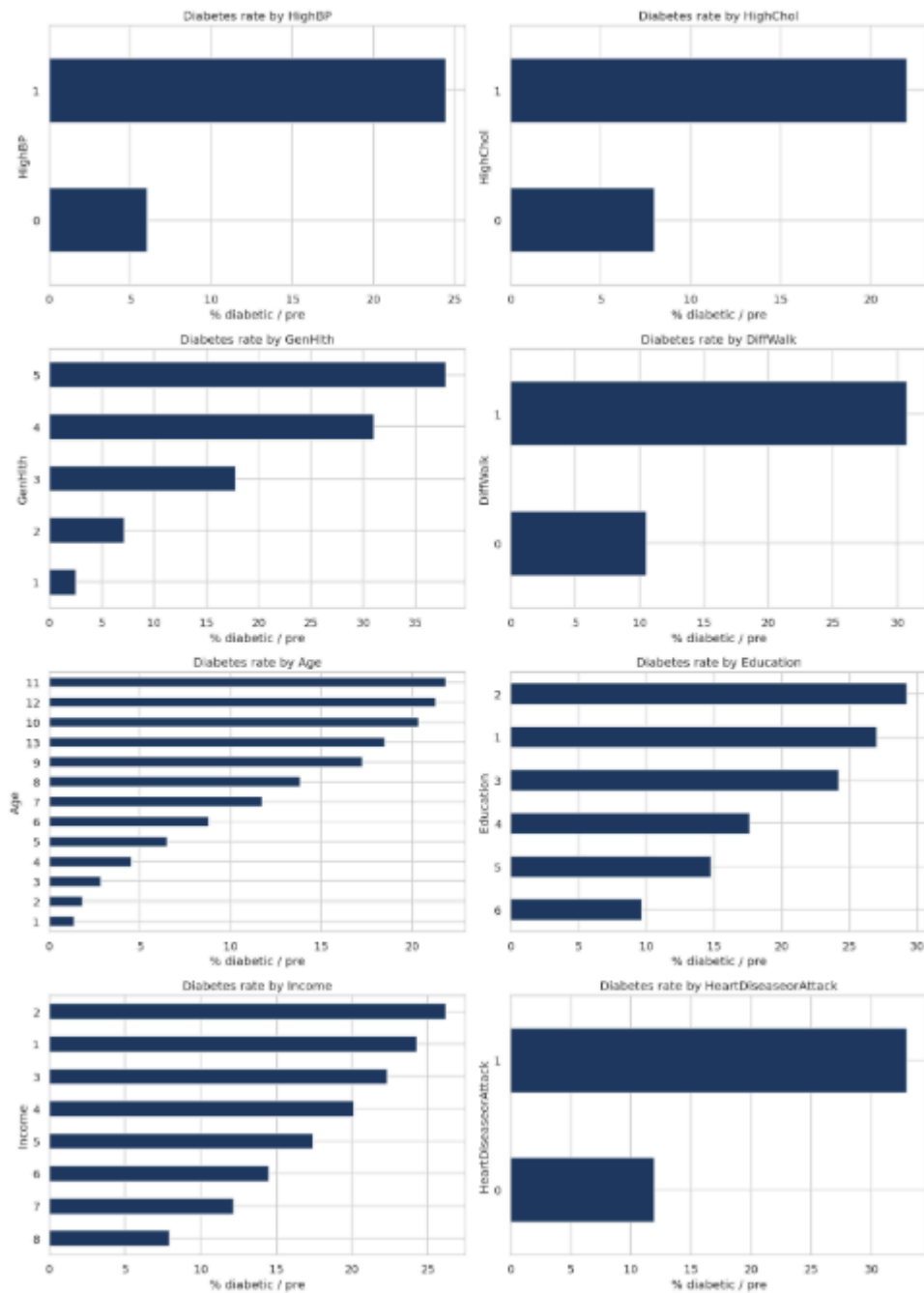


Figure 2. Rate of diabetes by key categorical and ordinal predictors. There is strong separation between general health, high blood pressure, high cholesterol and difficulty walking and there is a clear gradient across age, education and income.

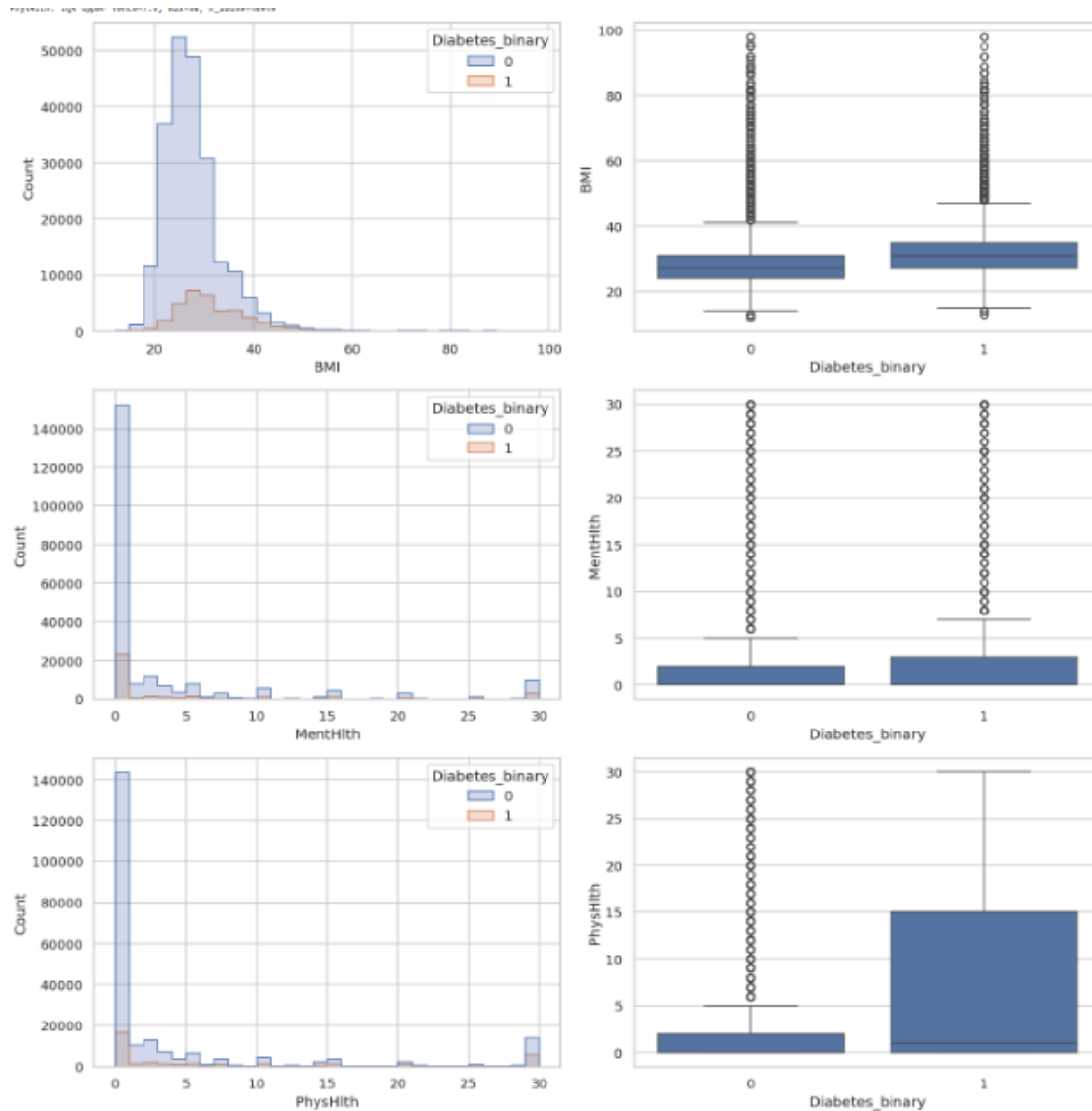


Figure 3. Distributions of the continuous features by class. Body-mass index is right skewed with real high outliers and mental- and physical-health day-counts are capped at 30.

3.2 The preprocessing and feature handling.

Before any train–test split, I removed all the 24,206 exact copies. This matters because if two question vectors with the same answer dropped on either side of the split, then the test set would contain some of the same questions as the training set and increase the scores. Before the removal of the duplicates, 13.9 percent of the records were positive, but after the removal, 15.3 percent were positive, leaving 229,474 records. The 9,820 were capped (not dropped) at the upper inter-quartile fence of 41.5 for body-mass index, which does not lose any of the sample size, but at the same time does not give excessive leverage to extreme values. In the mental- and physical-health fields, the construction penetrated at 30 days without needing to be capped.

The 21 predictors were split into two groups for encoding. Seven features were continuous or had a meaningful order, and were standardised to have mean 0 and variance 1, and included BMI, self rated general health, mental health days, physical health days, age band, education and income. The other 14 binary indicator flags were one-hot encoded. Both transformations were concatenated together and

a scikit-learn ColumnTransformer was fitted on the training partition only and then applied to validation and test data without modification, resulting in a leak-free pipeline and 35 input dimensions post-encoding. The data was stratified on the target and partitioned into training, validation and test sets of 64%, 16% and 20% respectively ensuring class balance in each partition.

3.3 Models and experiments.

I did 9 experiments in two model families, all of which I logged through the same harness so that they're all the same. The six classical experiments were logistic regression without and with balanced class weights; a simple random forest and a random forest with balanced class weight (300 trees, max depth 12, leaf size 5); a histogram-based gradient boosted tree; and a radial-basis support vector machine with class-balanced weights. The kernel support vector machine suffers from the following issues: it scales quadratically with the number of rows in the training set, it is impractical when the size of the training set is about 147,000, and it is useful to demonstrate what a margin-based learner learns from a representative subset of the data set.

A common tf.data pipeline was used in all three deep learning experiments for the purpose of batching the data into a set of 256 and shuffling the training stream. The first was a Sequential multilayer perceptron with two hidden layers (64 units and 32 units in the hidden layers, both with ReLU activation). The second was a Sequential network with a wider first layer, batch normalisation, two dropout layers (at 0.3) and balanced class weights, to see if explicit regularisation and class weight balancing would achieve improvements in the ranking. In the third, the planar architecture was implemented as a two-branch model with a ReLU and a tanh branch that processes the same input in parallel and are then combined before the output, to analyze if a deliberately heterogeneous mix of non-linear components is helpful on tabular data. All three employed Adam optimiser, early stopping on validation ROC-AUC with a patience of 6 epochs and weight restoration, being the main preventative measure from over-fitting. The training distribution used was translated into class weights of around 0.59 for the negative class and 3.27 for the positive class.

All the models were assessed for 6 parameters namely, accuracy, precision, recall, F1 score, ROC-AUC, and area under precision-recall curve. ROC-AUC, PR-AUC and threshold-independent to measure the ability to rank cases and precision, recall and F1 at a set threshold to measure the performance as a decision rule. In order to have a reproducible entire pipeline, a fixed random seed of 42 was defined for NumPy, Python, scikit-learn and TensorFlow.

4. Results.

All nine experiments from the validation set are listed in Table 1, in order of running. Most of the models are grouped tightly; 8 out of 9 models fall in a band of 0.80 to 0.82 between each other, and even the simplest model studied (logistic regression) is less than 0.007 behind the best neural network. The only obvious laggard is the default random forest with a score of 0.776, which improves to 0.813 when restricted to "depth" and "balanced classes".

Experiment	ROC-AUC	PR-AUC	F1	Recall	Precision	Acc.
E1 LogReg (no weight)	0.811	0.432	0.263	0.171	0.574	0.854
E2 LogReg (balanced)	0.812	0.431	0.453	0.750	0.324	0.722
E3 RandomForest (default)	0.776	0.360	0.262	0.181	0.472	0.844
E4 RandomForest (tuned+bal)	0.813	0.437	0.459	0.745	0.332	0.732
E5 HistGradientBoosting	0.817	0.448	0.250	0.158	0.605	0.855
E6 SVM (RBF bal, 20k subsample)	0.803	0.386	0.089	0.049	0.502	0.847
E7 TF Sequential (baseline)	0.817	0.442	0.249	0.158	0.597	0.855
E8 TF Sequential (BN+drop+CW)	0.813	0.435	0.445	0.796	0.309	0.696
E9 TF Functional (2-branch+CW)	0.818	0.445	0.454	0.777	0.321	0.714

Table 1. Validation results for all nine experiments. ROC-AUC and PR-AUC are threshold independent ranking measures whereas F1, Recall and Precision are at the default threshold of 0.5.

4.1 Classical results.

The classical models include two patterns. The first relates to class weighting. Moving logistic regression from unweighted (E1) to balanced (E2) only slightly improves the ranking quality (ROC-AUC from 0.811 to 0.812), but also changes the model as a decision rule: recall at the default threshold increases from 0.171 to 0.750, while precision decreases from 0.574 to 0.324. That is, when one talks about balancing, one is not talking about making the model better at separating cases, but is merely talking about moving the default cut-off point around in the distribution of scores. It's the same for random forest, recall increasing from 0.181 to 0.745 when balancing and depth tuning. The second is the lack of performance of the default random forest. Without restriction 0.776 is assigned to its deep trees as it is dominated by the majority class and without balancing the classes, it ranks the cases poorly; with depth limitation and the balancing of the classes, it shows a competitive rank of 0.813. The best classical model was histogram gradient boosting, with a score of 0.817, and the quickest of the high-capacity learners to converge on the entire data when the other models were still learning. This subsampled SVM achieves a respectable accuracy of 0.803 on a slice of the data, and is only included here to illustrate the accuracy/compute tradeoff; it's not a serious competitor.

4.2 Deep-learning results and learning curves.

In no material way, the deep models outperform or underperform the best classical models. The ROC-AUC of the Functional two-branch network is the highest of all the networks studied, but is still below 0.001 from histogram gradient boosting and the baseline Sequential network. This is a reliable result and not just a lucky run as will be seen from the learning curves in Figure 4 – training and validation ROC-AUC tracks closely throughout and early stopping stops each network before the two curves begin to diverge, meaning that the networks are not over-fitting. Interestingly, the capacity, batch normalisation, and dropout did not affect the ranking performance as the regularised Sequential

network (E8) did not outperform the plain baseline (E7), but rather shifted the recall–precision curve using the class weights, with a similar ROC-AUC.

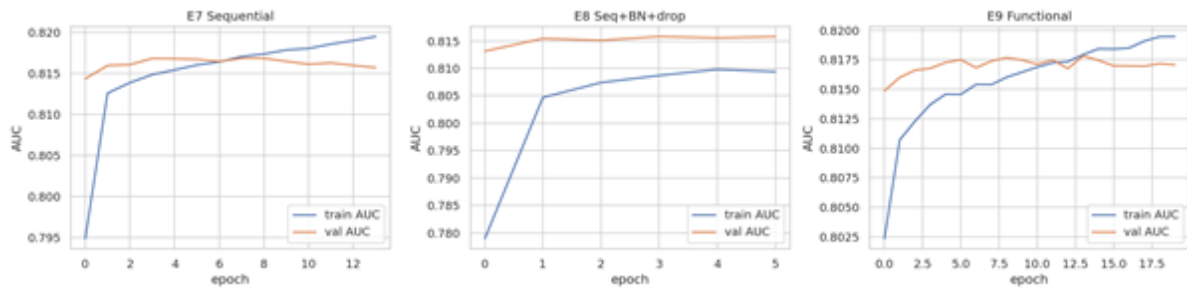


Figure 4. The three networks were trained and validated with the ROC-AUC method. The low variance and lack of over-fitting are evident by the close fit of the curves, and from the early stopping before any divergence.

Another near 100% tie in the study is the convergence of three widely different model families: a linear model, a boosted-tree ensemble and a neural network, all of which come close to the same ROC-AUC score; I make this point again in the discussion under the heading of the bias-variance interpretation.

4.3 Test-set performance is available, along with a threshold tuning function.

I took the last two finalists, histogram gradient boosting and baseline Sequential network, to the held-out test set. Their test ROC-AUC values are 0.818 and 0.819 respectively, which is a statistical tie that is the confirmation of the validation picture on unseen data. However, both models are poor decision rules at the default setting of 0.5; the network recalls only 17 percent of the true cases, even though it has a very high ranking, because under imbalance the default cut-off is too conservative. Changing the threshold used for validation data to the optimal F1 score (0.241) alters the scenario completely. The network's recall improves from 0.168 to 0.612 and F1 increases from 0.262 to 0.475, precision decreases from 0.603 to 0.388, but most importantly, ROC-AUC remains 0.819, as moving the threshold doesn't affect the underlying ranking. The ROC and precision–recall curves of the two finalists are plotted in Figures 5 and 6, respectively, as well as the confusion matrices prior to and after tuning.

Model / threshold	ROC-AUC	PR-AUC	F1	Recall	Precision
HistGradientBoosting @0.50	0.818	0.445	0.249	0.157	0.598
Neural net @0.50	0.819	0.444	0.262	0.168	0.603
Neural net @tuned (0.241)	0.819	0.444	0.475	0.612	0.388

Table 2. Two finalists were tested on the test set and the impact of changing the decision threshold of the network. The operating point, but not the ranking quality (ROC-AUC, PR-AUC) changes depending on the threshold.

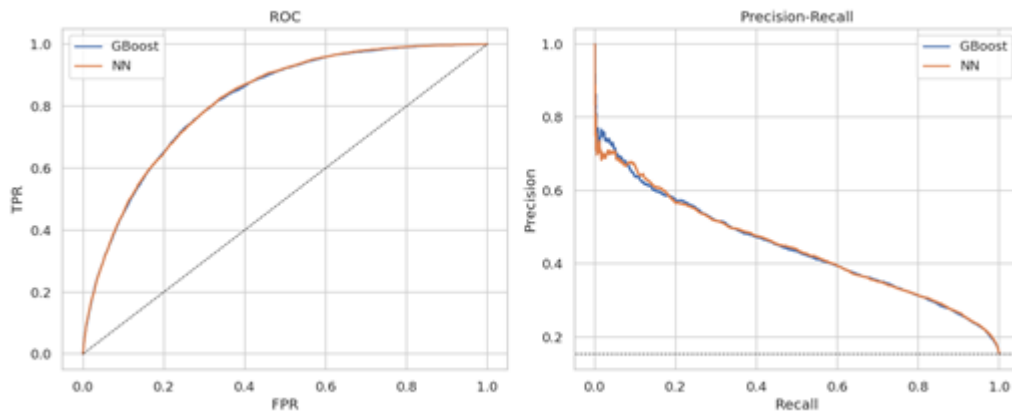


Figure 5. ROC and precision–recall curves of the two finalists on the test set. The curves are very similar, also consistent with the tied AUC values.

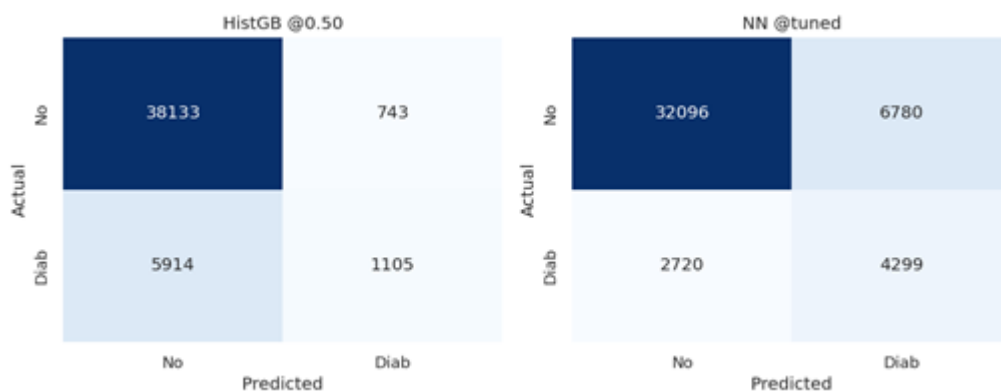


Figure 6. The left figure is a confusion matrix of gradient boosting trained using the default threshold, which fails to capture most of the true cases, while the right figure shows the network trained with the threshold that recovers the majority of the true cases while adding more false positives.

5. Error Analysis

5.1 How the model was developed.5.3 Understand its construction.

The gradient-boosting final model indicates that the most important feature is self-rated general health, followed by body-mass index, age, high blood pressure, and high cholesterol (as shown in Figure 7). This ordering is reasonably clinically sensible and in line with the exploratory analysis. All the same, it is important to consider the equity discussion that follows: the model is heavily dependent on general health, BMI and age, which in turn are all correlated with socio-economic status, meaning the distribution of the errors of the model is not socially neutral.

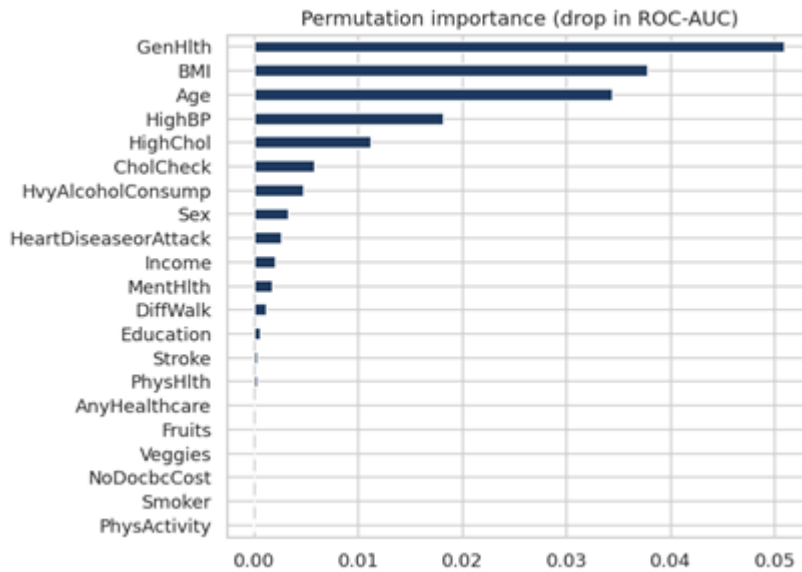


Figure 7. Permutation importance scores for the gradient-boosting model, which correspond to the loss of ROC-AUC on shuffling each feature. General health, body-mass index and age dominate.

5.2 Subgroup recall and the equity finding.

The key error analysis finding is that the tuned model's recall on true cases (the percentage of those who are actually diabetic who are recalled) varies according to subgroups. Gender disaggregation reveals almost no gender differences: the recall is 0.610 and 0.615, respectively, for the female and male groups, with the model being: virtually equal on this axis. The narrative of education and income is very different, and at first seems paradoxical. Decreases in recall gradually with both. The model recovers 83 percent of true positive observations in the lowest band of education but only 50 percent in the highest band of education; recall decreases from 0.770 to 0.422 respectively across the lowest and highest levels of education. The age pattern is the most drastic of all: the very youngest respondents have almost zero recall; the older bands have around 0.72; and the very oldest show a second dip in recall. These gradients are summarised in table 3.

Subgroup axis	Lowest band	→ mid	→ upper	Highest band	Pattern
Sex	0.610	—	—	0.615	No disparity
Education (1→6)	0.833	0.777	0.613	0.500	Recall falls as education rises
Income (1→8)	0.770	0.682	0.518	0.422	Recall falls as income rises
Age (1→13)	0.083	0.387	0.681	0.499	Rises with age, dips at oldest

Table 3. Recall on real diabetic patients within subgroups, at the tuned threshold. Selected bands are displayed; the income and education gradients are monotonic between the displayed bands.

This pattern is not a coding error, but instead is base-rate driven. Lower income and lower education groups have higher prevalence of diabetes, and the socio-economic gradient has been learnt by the model as a risk marker. It therefore tends to give high scores to low-status respondents, which means that it gives a high score to most of the cases that are truly present in that group; it also gives a low score to high-status respondents, meaning that it has a lower rate of recalling the cases that are actually present in that group. There are, then, two types of failure, however: Higher-income patients and higher-education patients have a higher likelihood of a false negative (meaning that they have true disease, but it is missed), and lower-income patients have a higher likelihood of a false positive (meaning that they do not have true disease, but they are referred anyway). Failure is not innocuous in either situation, and a single measure of accuracy (approximately 0.85) covers both. That's the same lesson that Obermeyer and others learned in another context: Heterogeneous behaviour across groups can be aggregated and still be materially different [11]. They actually aren't bigger or more complicated, it's just group-aware thresholds on the decision or post-hoc calibration that matches the error profile across bands.

6. Discussion

Based on the results four conclusions are drawn, one of which is the fourth one on equity. The one which is classical versus the deep is the first. For this tabular health data, deep learning did not provide any benefits. The best network was within 1 part in 1000 of gradient boosting in terms of ROC-AUC and a regular logistic regression was within 7 parts per thousand. This is precisely what the literature on tabular learning says [4, 5, 6]: If the predictors are heterogeneous, low cardinality and individually informative, the inductive biases of the tree ensemble are well suited to the problem and the additional capacity of a neural network is not much. Since gradient boosting is faster to train, less parameters to tune and directly interpretable feature importances, it is the pragmatic deployment choice. The neural networks were trained as a nice close approximation and were not suggested as the model, but rather used as a solid control.

The second conclusion has to do with the type of challenge. The biggest of the problems here is class imbalance; class imbalance is a problem here, not capacity. The evidence is clear: the class weighting had a huge impact on the recall, whereas the ROC-AUC was hardly affected; the decision threshold was increased, raising the recall from 0.17 to 0.61, but the results were the same, i.e., no changes in ranking. But the information to rank cases was there, and what was added was only an operating point – the more capacity the more models you would have found that gain. This redefines the engineering problem to one of setting a threshold that is meaningful in the real-world context of the screening situation, with regard to the costs of missed tests and false alarms.

The third conclusion is the bias–variance version of the convergence in Table 1. If a linear model, a boosted-tree ensemble and a neural network all achieve 0.82 ROC-AUC, it is very unlikely that the limitation is variance or capacity, because these three families of models have very different limitations in terms of variance and capacity. The more likely reason is that the issue is bias-limited: The information collected in these 21 self-reported survey sets sets a ceiling on what a model can achieve at separating the classes, and all three families are close to that ceiling. This is further supported by the learning curves shown in Figure 4 which are all very similar with little variance and not over-fitting. It's suggesting that there will be gains in the future from richer features, such as lab values and longitudinal values, and not more complicated models.

The last and most crucial finding is the equity finding. The errors from the model are not homoscedastic, due to the general health, BMI, age and SEG. The association between recall on true cases and income and education gets weaker with increasing income and education; thus, one global threshold systematically misclassifies more genuine cases among the more advantaged respondents and more disadvantaged cases among the more disadvantaged respondents. This is very important for a tool for which the sole purpose is to focus where scarce screening resources are needed, and this wouldn't be seen without the subgroup analysis.

7. Restrictions and Further work.

There were a number of constraints on these conclusions. Data is self-reported from a telephone survey conducted in the United States in 2015 and therefore subject to recall and selection bias and is not clinically confirmed; label indicates whether respondent was instructed by a physician that they have diabetes, when this is a mix of true diabetes and access to diagnosis and is the reason for the socio-economic error gradient. There are no laboratory values (glycated haemoglobin or fasting glucose) in the data set that have the greatest potential to bring performance over the apparent 0.82 ceiling. The binary target combines pre-diabetes and diabetes, and the survey is cross-sectional in design, which means that it can't provide any temporal or causal inferences. Lastly, the findings are limited to the United States, and the magnitude of the socio-economic error gradient, and even its form, would vary in other African countries or globally if the income/education/access-to-diagnosis relationship was different. It should be interpreted, therefore, as a methodological template, screening heuristic, and not as a diagnostic tool for transfer.

The following are limitations that are direct consequences of the work that is to be done in the future. The best thing to do is to replace the single global threshold with group-aware thresholds or post-hoc calibration, where the threshold is set according to the costs of missed cases and false alarms in a given screening programme and not its maximised F1 score in the abstract. In addition, the validation of the approach on African cohort data; the inclusion of affordable laboratory or mobile collected measurements; and the use of a three-class version of the data, separating pre-diabetes from diabetes would further strengthen the work and help bring it closer to a deployable tool.

8. Conclusion

In this project, we have examined the prediction of diabetes using a large-scale, imbalanced BRFSS dataset with a single leak-free data pipeline by applying classical machine learning and deep learning methods to nine experiments. The main takeaway from the findings is that the model family choice makes almost no difference on this type of tabular data: a linear model, a gradient-boosted ensemble and two NN models all resulted in a ROC-AUC of around 0.82 and gradient boosting is the practical and go/no-go model for this type of data. What is important, however, is how to deal with class imbalance, which was the problem that was found to be a threshold problem, a problem that was addressed not by adding more capacity to the model, but by class weighting and tuning the threshold. Most importantly, a subgroup error analysis showed that the accuracy of the model decreases with increasing income and education, which was not apparent in the overall accuracy, and is crucial to using such a model in a responsible way as a screening tool. The engineering challenge here is to refine and rethink thresholds, not larger models.

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